An Intelligent E-Learning System for Autistic Children: Multi-Agent Architecture

Najoua Tahiri(✉) and Mohamed El Alami

National School of Applied Sciences of Tangier, Abdelmalek Essaadi University, Tangier, Morocco
najoua.tahiri@gmail.com, elalami-hassouni@ensat.ac.ma

Abstract. E-learning systems are the result of the new era of digitalization, which has affected many areas including education. These systems offer many benefits like ease of access and time management. Therefore, the educational paradigm has been changed. Otherwise, since E-learning systems consider only the cognitive state of the learner regardless of his emotional state, they cannot instruct students effectively. Moreover, if the target learner has non-ordinary cognitive and socio-emotional abilities, such as a child with ASD (Autism Spectrum Disorder). So, the e-learning system must be able to satisfy those needs and propose a suitable content according to his learning rhythm.

This paper proposes a new multi-agent architecture of an emotional intelligent e-learning system, it aims to help ASD children overcome learning impairments. The proposed architecture is based on several agents, endowing to address emotional, cognitive and pedagogical issues intelligently. They are working in a collaborative and cooperative fashion to provide the appropriate content.

Keywords: Multi-agent architecture · E-learning · Intelligent learning system · Affective computing · Autism

1 Introduction

In recent years, the educational paradigm has been changed due to the arrival of E-learning concept. It stems from the ongoing incorporation of technology into learning and education environments. For Som [1]: E-learning refers to the systematic use of networked information and communications technology in teaching and learning. Also, Henry [2] defines E-learning as the appropriate application of the Internet to support the delivery of learning, skills and knowledge in a holistic approach not limited to any particular courses, technologies, or infrastructures. Similarly, Jenkins et al. describe E-learning as learning facilitated and supported through the utilization of information and communication technologies (ICTs) [3]. In brief, there are several definitions of E-learning but all definitions overall describe the use of electronic tools in the process of learning. However, these systems adopt the one-size-fits-all strategy of delivering content to a passive learner. They have undergone several changes from computer-based training program until the appearance of the artificial intelligence. Consequently, a new generation of Intelligent E-learning Systems (ILS) has started to overcome challenges of classical E-learning systems [4, 5].

© Springer Nature Switzerland AG 2020
M. Ezziyyani (Ed.): AI2SD 2019, AISC 1102, pp. 83–90, 2020.
https://doi.org/10.1007/978-3-030-36653-7_8

ILS derives from an approach known as ITS (intelligent tutoring system) which has been pursued by researchers in education, psychology, and artificial intelligence. The goal of ITS is to provide the benefits of one-to-one instruction. It enables learners to practice their skills by carrying out tasks within highly interactive learning environments [6]. While establishing better emotional interaction with the learner through an intelligent E-learning system requires interpreting his socio-emotional reactions in order to provide a personalized instruction adapted to his knowledge, learning abilities and his needs. Existing intelligent E-learning systems support only the cognitive aspect of the learner by neglecting his emotional state.

Many research studies in neuroscience and psychology show the relevance of emotions in the learning process as a substrate that can promote a better development of cognitive functions. Damasio [7] has elucidated the neural bases of emotions as well as the strong link between emotion and certain cognitive processes such as attention, memory, perception and decision-making.

Including emotions in such computerized process require incorporation of Affective Computing. It is a field of interdisciplinary research that involves studying the interaction between technology and feelings, including giving machines the ability to recognize, express, synthesize and model emotions [8].

Emotionally aware systems are regarded as a considerable and valuable educational technique particularly for children with special cognitive needs such as ASD children who do not respond naturally to linear curricular delivery, but rather need an educational model incorporating intra and interpersonal development skills.

In this perspective, the article first draws up the related work to ILS based on multi-agent architecture. Then, it establishes the proposed architecture of an intelligent e-learning system for autistic children. Third, it describes a discussion and future work section. Conclusions are presented in the last section of this paper.

2 Related Work

Affective Computing is computing that relates to, arises from, or deliberately influences emotion or other affective phenomena [8]. It is an interdisciplinary field covering computer science, psychology and cognitive science. In its simplest form, emotional learning characterizes the emotional area of learning reflected in the beliefs, values, interests, and behaviors of learners [9]. Emotional learning is concerned with how learners feel while they are learning, as well as how learning experiences are internalized so that they can guide learner attitudes, opinions, and behavior [10]. The main application of affective computing includes emotional perception, emotional understanding and emotional expression.

Many research studies are related to intelligent E-learning systems, some are based on the multi-agent architecture (MAS), while others are still using the traditional software architecture. The common goal of all these systems is to foster learning by ensuring some features such as interactivity, adaptability and collaboration.

During a process of learning, more the learner is interested and attracted by the training, more his concentration increases and vice versa. Hence, the learner's behavior changes according to his feelings. The authors [11] propose an E-learning system model is composed of two agents: learner and teacher, it integrates a new component called affective computing module. Learner's emotions are collected through facial expressions and analyzed at this level. Afterward, the teacher agent change the pedagogical content based on result of emotional module in order to regulate learner emotional state.

Moreover, BOKHARI-Intelligent Multi-agent based E-learning System [12] is built on a four layers: user level, web level, system level and data level. The focus of the BOKHARI system is to incorporate many features like intelligence, accessibility, interactivity into a single system. Also, Daomin describes in his paper [13] a multi-agent based E-learning system MAELITS applying the methodology of case-based reasoning. This system selects and deliver appropriate oriented object learning courseware based on SCORM and according to student's cognitive ability for individual student. Four agents (courseware interface, examination, courseware recommendation, and activity data collection) interact with E-learning curriculum creation model that is responsible on delivering the appropriate pedagogical content. However, these two systems (BOKHARI and MAELITS) do not include the emotional state of the learner.

About intelligent learning systems based on MAS architecture for autistic children, we quote the application of an agent-based approach proposed by Sehaba et al. [14], it aims to the rehabilitation of children with autism, where the multi-agent system represents the child's profile, the expert's directives and the dynamics of their interactions analyze of child's behavior in order to adapt the activities execution.

Also, Barakova et al. [15] have developed an interactive multi-agent platform of 15 blocks with emergent behavior. It consists of building blocks that interact when positioned in each other's vicinity. These blocks can interact by detecting their neighbors through the built-in sensors, and show behavior in response to their sensory stimulation.

Despite the burgeoning need to consider new ways of providing efficient educational services, which adopt one-to-one teaching techniques for ASD children, we noticed that intelligent learning systems based on multi-agent architecture are very rare.

3 Proposed Work

This work proposes the architecture of an intelligent learning system for ASD children. It's spin-off NEOCAMPUS2 [16], an agent factory that helps designing and implementing intelligent agents. It directly provides a modular agent architecture, quite complete which if required, other modules could be easily added. On the other hand, it is the only one that facilitates affective and fuzzy computation. It also offers extensive built-in possibilities for cooperation and control of agents.

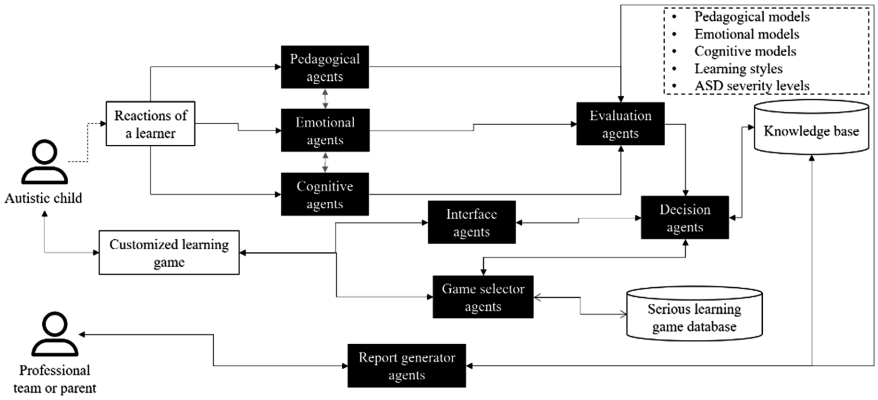


Fig. 1. Proposed system architecture

NEOCAMPUS [17] is a platform for the design and implementation of intelligent agents, with an extensive history in the production of very diverse prototypes, fundamentally, in the field of intelligent e-learning systems. Among them, we can quote MEDIC, FINANCE, FILTER, AFFECTION and ENT. MEDIC [18] is an intelligent e-learning system simulating the functioning of a medical center: its departments, finance, and managerial problems. A set of junior managers can learn to solve the different problems or incidents happening in the center daily life. FINANCE [19] is another intelligent e-learning system for learning financial or managerial accountant and firm analysis. Also, Diautis a multi-agent platform [20] that offers a wide and variable collection of tests to diagnosis ASD children, including computerized interviews with parents, using sensors to analyze the interaction of the child through the monitoring of ocular activity, the observation of movements, as well as the interpretation of vocal responses.

So, the proposed architecture benefits from NEACAMPUS's capabilities and functionalities, including affective computing and the use of pedagogic agents.

The main objective of this work is to foster learning by taking into account the ASD learner's cognitive, emotional and pedagogical specifications. Accordingly, we chose multi-agent based approach. The type of proposed architecture is heterogeneous communicating MAS where agents are of different models and granularity. The communication between agents stands on cooperation to break down the problems into sub problems and solve them more easily.

Several agent architectures used different representation approaches for their reasoning mechanism and propose different types of mental attitudes in order to solve a variety of problems. Particulary, the most adopted is the BDI (beliefs, desires, and intentions) architecture, originally proposed by Bratman [21] as a philosophical theory of the practical reasoning. It is a deliberative agent architecture where actions are derived from a practical reasoning process composed of two steps:

1. The deliberation (of goals), a set of desires is selected to be achieved, according to the current situation of the agent's beliefs.

2. The determination of how these concrete goals produced as a result of the previous step can be achieved by means of the available options for the agent [22].

Since BDI architecture provides a description of behavior at a much higher level, in terms of mental attitudes. It allows agents to be more autonomous and proactive. The straightforward connection between BDI architecture and emotional abilities constitutes an argument in favor of using BDI agents in simulations where emotional abilities are needed such as the proposed architecture.

The suggested architecture incorporates serious learning games as pedagogical activities that insure the child's engagement; it allows teaching pedagogical concepts implicitly through games and receiving the perception of the child in the reverse circuit.

And in order to provide the appropriate serious learning games according to the emotional, cognitive, pedagogical and sensory aspects of the child learner, the authors propose to use the KASP [23] methodology to allow a better design of the games. It is a new methodology of designing serious learning games for children with learning disabilities in order to improve their learning rhythm and improve their emotional state related to learning in a playful and interactive environment.

Figure 1 shows the general architecture of the system. It includes several agents:

- ***Emotional agent***: It constantly analyzes the learner's emotional state along the proposed pedagogical activities using the information provided by the sensors. It communicates with the cognitive agents to obtain the necessary inputs for the analysis of the emotional state of the learner.
- ***Cognitive agent***: Cooperating with emotional agents to analyze the learner's reactions of cognitive origin, they measure the learner's ability to learn a specific concept.
- ***Pedagogic agent***: Receive instructions from emotional and cognitive agents to accompany the learner in learning sessions by promoting his interest in the learning object. Following the data collected through sensors, it identifies pedagogical gaps of the child.
- ***Interface agent***: It is the communication bridge between the user and the system. It proposes a personalized content considering the actual situation of the learner. It collaborates with game selector agent in order to customize game interface according orders getting from decision agent.
- ***Evaluation agent***: Responsible for obtaining the evaluation of the learner's reactions using the inputs provided by the cognitive, emotional and pedagogic agents; Also responsible for updating and storing user's models in knowledge base.
- ***Decision agent***: It establishes new orders for game selector agents and interface agents based on evaluation agent requests and recommendations.
- ***Game selector agent***: It searches for nearby scenarios in the serious learning game database. For each particular situation, game selector agents may react differently by using the inputs supplied by decision agents.
- ***Report generator agent***: It allows monitoring the child's progress through reports generated for parents or instructors in order to recognize behavioral, socio-emotional state and offer a parallel support in his daily life.

The agents are based on BDI (beliefs, desires, and intentions) architecture. The BDI approach has been proposed in Artificial Intelligence to represent the way agents can do complex reasoning. The proposed architecture is used to monitor the learning process of an ASD child through an intelligent E-learning system that takes into account three aspects: cognitive, emotional, and pedagogical.

The proposed architecture is based on MAS, and benefits from the potential of cooperation, which is an effective way of communication between agents, and the capacity of learning from children's interactions as well. Thus, it allows to build a system for different types of people revolving around autistic children, which are parents, clinical staff that includes speech therapists, psychiatrists, pediatricians and ASD specialists (applied behavior analysts), and tutors such as teaching assistants, support assistants and educational professionals.

4 Future Work

The proposed architecture could be used not only for ASD children but it can also serve children with learning disabilities since autism and learning disability are co-associated. According to the NHS Information Centre, Community and Mental Health Team [24], between 31 and 35.4% of people with a learning disability are autistic. It will also improve diagnosis of ASD children through the acquisition of children's mental models.

As part of the continuity of this work, we attempt to develop an exhaustive platform that includes a system based on the current architecture and helps ASD children as a coach in their learning, from diagnostic stage until the progress monitoring, offering a computerized psychological tools and a personalized pedagogical activity by way of serious learning games. The future platform will also offer effective tools intended to assist pedagogical and medical team revolving around ASD children.

5 Conclusion

The learning environment has a significant impact on children with ASD. Careful consideration should be given to the physical, sensory and social environment when supporting them to access learning opportunities. The proposed architecture is based on the multi-agent approach and integrates affective computing as one of its main components. Each agent treats a specific area of knowledge and collaborates with other agents to provide instructional content appropriate to the learning style as well as the emotional state of the ASD child. Furthermore, multi-agent architecture presents a better approach to provide a collaborative environment; it offers several advantages, among them the scalability that allows equipping the system with new capabilities by simply adding new agents.

References

1. Naidu, S.: E-Learning: A Guidebook of Principles, Procedures and Practices, 2nd edn, 88. P. Commonwealth Educational Media Center for Asia (CEMCA), New Delhi, India (2006)
2. Paul, H.: Elearning Technology, Content and Services, Education + Training **43**(4), 249–255 (2001)
3. Jenkins, M., Hanson, J.: E-learning Series: A Guide for Senior Managers, Learning and Teaching Support Network (LSTN). Generic Centre, United Kingdom (2003)
4. de Arriaga, F.:. E-Knowledge Management, ELearning and E-Commerce: An Evaluation of Their Situation and Tendencies. International Computer Science Institute, Technical Report, University of California, Berkeley, pp. 1–56 (2003)
5. de Arriaga, F., El Alami, M., Arriaga, A.:. ELearning: new generation of intelligent web based systems. In: Proceedings 14th International Conference on Computer Theory and Applications, pp. 191–198. Alexandria (2004)
6. Phobun, P., Jiracha, V.: Adaptive intelligent tutoring systems for e-learning systems. Procedia Soc. Behav. Sci. **2**, 4064–4069 (2010)
7. Damasio, A.: Descartes Error: Emotion. Reason and the Human Brain, Vintage Digital, London (2008)
8. Picard, R.W.: Affective Computing. The MIT Press, Cambridge, MA, US (1997)
9. Gano-Phillips, S.: Affective learning in general education. Spec. Top. Assess. Univ. General Educ. Program **6**(1), 1–44 (2009)
10. Krathwohl, D.R., Bloom, B.S., Masia, Bertram B.: Taxonomy of Educational Objectives: The Classification of Educational Goals. Handbook II: Affective Domain. McKay, New York (1964)
11. Duo, S., Xue, L.S: An e-learning system based on affective computing. Phys. Procedia **24** (2010). https://doi.org/10.1109/icime.2010.5477869
12. Bokhari, M.U., Sadaf, A.: BOKHARI-Intelligent Multi-agent based e-Learning System (IMBLS) for Interactive Distance Learning (2014)
13. Daomin, X.: Multi-agent based e-learning intelligent tutoring system for supporting adaptive learning. In: Proceedings—2013 4th International Conference on Intelligent Systems Design and Engineering Applications, ISDEA 2013, pp. 393–397 (2013). https://doi.org/10.1109/isdea.2013.494
14. Sehaba, K., Estraillier, P.: A Multi-Agent System for Rehabilitation of Children with Autism (2005)
15. Barakova, E., Gillessen, J., Feijs, L.: Social training of autistic children with interactive intelligent agents. J. Integr. Neurosci. **8**, 23–34 (2009). https://doi.org/10.1142/s0219635209002046, https://elearning.autism.net/
16. de Arriaga, F., El Alami, M.: NEOCAMPUS2: A multi-agent environment for educational research and applications. In: Innovation, Technology and Research in Education. Iadat (2004)
17. de Arriaga, F., El Alami, M., Arriaga J., Arriaga, F.: NEOCAMPUS: multi-agent software environment for on-line learning. In: Proceedings of International Conference on Information Technology and Education, ICT'2002, pp 1245–1255. Badajoz (2002)
18. de Arriaga, F., El Alami, M.: MEDIC2: Evaluation of a Fuzzy Intelligent Learning System (2006)
19. de Arriaga, F., El Alami, M., Arriaga, A.: Evaluation of Fuzzy Intelligent Learning Systems Recent Research Developments in Learning Technologies (2005)

20. El Alami, M., Tahiri, N., de Arriaga, F.: DIAUTIS: a fuzzy and affective multi-agent platform for the diagnosis of autism. Br. J. Appl. Sci. Technol. **21**, 1–28 (2017). https://doi.org/10.9734/BJAST/2017/33716
21. Bratman, M.E.: Intentions, Plans, and Practical Reason. Harvard University Press, Cambridge, MA (1987)
22. Wooldridge, M.: An Introduction to Multiagent Systems, 2nd edn. John Wiley & Sons Inc, New York (2009)
23. Tahiri, N., El Alami, M.: KASP: A cognitive-affective methodology for designing serious learning games. Int. J. Adv. Comput. Sci. Appl. **9** (2018). https://doi.org/10.14569/ijacsa.2018.0911103
24. Brugha, T. et al.: The NHS Information Centre: Community and Mental Health Team: Estimating the prevalence of autism spectrum conditions in adults: extending the 2007 Adult Psychiatric Morbidity Survey. Leeds: NHS Information Centre for Health and Social Care (2012)